Assessing the Evidence for AGI Designations: A Formal Mathematical, Epistemological, and Empirical Audit of GPT-6 Astra

Technical Audit Series (Ref: EAG-2026-ASTRA-01) · September 8, 2026

Primary Materials Examined: OpenAI GPT-6 Astra System Card (September 3, 2026); OpenAI GPT-6 Astra Launch Disclosures; ARC Prize Technical Report on GPT-6 Astra; ICLR 2026 & Medical Foundation Model Technical Releases.

Abstract

This paper delivers a rigorous mathematical, epistemological, and empirical audit of claims declaring OpenAI's **GPT-6 Astra** (released September 3, 2026) as having achieved Artificial General Intelligence (AGI). Evaluating public disclosures through algorithmic information theory, statistical learning bounds, and formal computability limits, we establish that fixed-parameter autoregressive models fundamentally operate as high-dimensional density estimators bounded by $\mathrm{SPACE}\big(O(N)\big) \subsetneq \mathcal{R}$, rendering them structurally incapable of universal generalized reasoning.

Empirically, we analyze harness configuration sensitivity on ARC-AGI-3—revealing a 35.9-percentage-point performance gap between standardized (62.7%) and proprietary adapter interfaces (98.6%)—and audit cross-domain portability across technical evaluation suites. We evaluate these results against OpenAI's Charter definition (*"highly autonomous systems that outperform humans at most economically valuable work"*) and present a formalized, Popperian falsifiability matrix. We conclude that Astra's AGI designation remains **unverified on the record**, with the burden of substantiation remaining strictly on the claimant.

1. Introduction & Epistemological Demarcation

Commercial assertions of Artificial General Intelligence (AGI) routinely extrapolate high benchmark scores on specialized evaluation suites to claim domain-general cognitive mastery. From the standpoint of formal epistemology and the philosophy of science, an unverified AGI announcement carries zero scientific weight as an established empirical finding unless it satisfies clear demarcation and operational criteria.

1.1 Occam's Razor and Solomonoff Induction

In formal inductive inference, Occam's Razor is mathematically formalized via Solomonoff's universal prior distribution $P(M) = 2^{-K(M)}$, where $K(M)$ denotes the Kolmogorov complexity of hypothesis M. When observing high-watermark outputs on specialized benchmark tasks, two competing hypotheses exist:

- **Hypothesis H_0 (Statistical Manifold Interpolation):** The system is a finite-parameter model $M_\theta: \mathcal{X} \to \mathcal{Y}$ executing conditional density estimation $\hat{P}(Y \mid X)$ over the manifold defined by its training distribution support $\mathcal{D}_{\text{train}}$.
- **Hypothesis H_1 (Universal Generalized Intelligence):** The system contains an unbounded causal reasoning engine capable of domain-agnostic program synthesis across arbitrary, un-encountered problem spaces $\mathcal{E} \in \mathcal{U}$.

Because $K(H_0) = K(\theta) + O(1) \ll K(H_1)$, Solomonoff induction requires assigning exponentially greater prior probability to statistical manifold interpolation (H_0) over generalized intelligence (H_1). Inferring H_1 without explicit, auditable proof violates Occam's Razor.

1.2 Epistemic Defaults and Burden of Proof

In formal logic, the rational baseline regarding an unverified extraordinary claim is non-acceptance (Status $=$ Unverified). Withholding acceptance of an AGI claim due to inadequate evidence assigns it the status of unproven ($\neg$Established), not false ($\neg$Capability). A critic is not required to empirically disprove an unverified claim; rather, the burden of substantiation rests entirely on the claimant.

1.3 Disentangling Falsifiability, Access, and Safeguards

To evaluate claims without internal weight access, three concepts must be strictly separated:

- **Logical Falsifiability:** A property of a proposition's logical structure. A claim is falsifiable if there exists a conceivable, non-empty set of empirical observations that would contradict it. A claim becomes unfalsifiable if its boundary conditions fluidly shift to re-explain any observed failure.
- **Practical Access Constraints:** Closed API permissions, hidden state mechanisms, and non-released model weights represent operational barriers to external testing, not intrinsic logical flaws in the hypothesis.
- **Methodological Safeguards:** Protocols such as pre-registering evaluation criteria and establishing explicit failure thresholds prior to testing are valuable choices that strengthen evidentiary weight.

2. Mathematical Bounds on Model Generalizability

To evaluate whether a static, fixed-parameter architecture M_θ can satisfy universal generalized intelligence, we formalize the structural constraints imposed by algorithmic complexity, statistical learning theory, and computability theory.

Theorem 1 (Information-Theoretic Prior Dominance)

Let $D = \{(x_i, y_i)\}_{i=1}^{n}$ be the observed benchmark performance data. The posterior odds ratio between statistical manifold interpolation (H_0) and universal general intelligence (H_1) satisfies:

$$\frac{P(H_0 \mid D)}{P(H_1 \mid D)} = \frac{P(D \mid H_0)}{P(D \mid H_1)} \cdot 2^{K(H_1)-K(H_0)} \gg 1$$

Because $K(H_1) \gg K(H_0)$, the posterior probability is dominated by the algorithmic complexity gap, proving that statistical interpolation is exponentially more probable given static benchmark evidence.

2.1 No Free Lunch Theorem and Domain Portability

By the No Free Lunch (NFL) theorem for supervised learning, for any two learning algorithms A_1 and A_2 evaluated over a finite input space $\mathcal{X}$ and output space $\mathcal{Y}$:

$$\sum_{f \in \mathcal{Y}^{\mathcal{X}}} \mathbb{E}_{X \sim \mathcal{U}} \left[L\left(A_1(D_f), f(X) \right) \right] = \sum_{f \in \mathcal{Y}^{\mathcal{X}}} \mathbb{E}_{X \sim \mathcal{U}} \left[L\left(A_2(D_f), f(X) \right) \right]$$

A static parameterization θ optimized on dataset $\mathcal{D}_{\text{train}}$ achieves a loss reduction on functions aligned with $\mathcal{D}_{\text{train}}$ at the mathematical cost of equal performance degradation on the uniform complement $\mathcal{Y}^{\mathcal{X}} \setminus \mathcal{D}_{\text{train}}$. Universal generalized intelligence requires a non-zero performance advantage across out-of-distribution, non-stationary environments. Because average error across all unobserved function spaces is invariant, no static parameter set M_θ can possess universal general intelligence.

Theorem 2 (Automata Memory Space Bound)

Let M_θ be an autoregressive model with d parameters, precision B bits, and maximum context length L. The total state space capacity is bounded by $N = (d \cdot B) + (L \cdot B)$ bits. The system is structurally equivalent to a Deterministic Finite Automaton (DFA) with state space $|\mathcal{S}| \leq 2^N$.

$$\text{Lang}(M_\theta) \subseteq \text{SPACE}\big(O(N)\big) \subset \text{PSPACE} \subsetneq \mathcal{R}$$

Because Universal General Intelligence requires computing partial recursive functions $g \in \mathcal{R}$ requiring unbounded tape SPACE(∞), M_θ is mathematically incapable of solving computable problems requiring memory scaling beyond $O(N)$.

2.2 Diagonalization Proof of Capacity Limits

Let $\mathcal{F}_{\text{comp}}$ be the countable set of all computable functions $f: \mathbb{N} \to \{0,1\}$. Assume for contradiction that a fixed architecture M_θ possesses generalized intelligence capable of computing any $f \in \mathcal{F}_{\text{comp}}$ via prompt encoding $p \in \mathbb{N}$. Define $g(p, x) = M_\theta(p, x) \in \{0,1\}$. Construct the diagonal function $d(k) = 1 - g(k, k)$.

By construction, $d(k)$ is a computable algorithm. Suppose there exists p_d such that $g(p_d, x) = d(x)$ for all x. Evaluating at $x = p_d$ yields $g(p_d, p_d) = 1 - g(p_d, p_d)$, giving $0 = 1$, a contradiction. Thus, M_θ cannot compute all generalized tasks.

3. Empirical Audit of GPT-6 Astra Disclosures

Evaluating the empirical record published alongside the September 3, 2026 launch of GPT-6 Astra reveals clear harness dependency, domain variation, and configuration sensitivity.

3.1 Environment Harness Dependency in ARC-AGI-3

OpenAI highlighted a top-line score of **99.9% on ARC-AGI-3** under high reasoning settings. Technical audits by the ARC Prize team (Kamradt, Chollet, & Knoop) revealed significant harness sensitivity:

Evaluation Configuration	**Reasoning Effort Setting**	**Reported Score**	**Harness Mechanics**
Standard Interface (Neutral)	Maximum Reasoning	**62.7%**	Model manages own context notes.
Provider Adapter (OpenAI)	Maximum Reasoning	**98.6%**	Automated state compaction.
Provider Adapter (OpenAI)	High Reasoning	**99.9%**	Full adapter scaffolding enabled.

At identical maximum reasoning effort, removing OpenAI's Provider Adapter results in a **35.9 percentage point drop** (98.6% down to 62.7%). This gap demonstrates that top-line performance relies heavily on external context-management scaffolding. Highlighting 99.9% while omitting the 62.7% neutral interface score introduces a risk of selection bias.

3.2 Evaluated Domain Performance & Portability Analysis

Evaluating Astra's reported scores across system card disclosures shows high domain variation:

Evaluation Area	**Benchmark Suite**	**Reported Metric**	**Operational & Evaluative Context**
Advanced Mathematics	FrontierMath Tier 4	**97.6%**	Closed-domain multi-step formal math.
Cybersecurity Exploitation	ExploitBench	**100.0%**	Full coverage across V8 engine vulnerabilities.
Recent Cyber Exploits	ExploitBench (June–Aug 2026)	**39.0% / 31.7%**	Post-cutoff vulnerabilities; found 2 zero-days.
Reverse Engineering	SRE-Bench	**99.2%** (Pass@4)	Binary reverse engineering without source code.
Agentic Software Tasks	Agents' Last Exam	**59.3%**	Reported score on software workflows.
Terminal Analysis	Terminal-Bench 4.0	**57.9%**	Software engineering and terminal tasks.
Workflow Automation	AutomationBench	**41.4%**	Multi-step enterprise task execution.
Internal Debugging	Internal Research Debugging	**78.05%**	Disclosed as below internal "High" target.

Evaluation Area	Benchmark Suite	Reported Metric	Operational & Evaluative Context
Offensive Cyber (Irregular)	FrontierCyber	**86 / 226 Solved**	0/7 Elite challenges solved on hardened targets.

Methodological Rigor on Score Parsing: Benchmark scores (e.g., 59.3% on *Agents' Last Exam*) must be interpreted strictly using each suite's scoring rules; they cannot be converted into binary "task failure rates" without verifying unweighted binary mechanics. Furthermore, percentage variations across uncalibrated suites reflect differing task difficulty baselines, not a controlled measure of transfer failure.

3.3 Differentiating Non-Examinable Procedures vs. Independent Audits

To maintain analytical precision, we differentiate third-party audited findings from non-examinable areas:

- **Independently Examined:** External testing by UK AISI (30.9-min No-CoT math time horizon), Apollo Research (0.17% label falsification rate; 50.6% evaluation awareness), SecureBio (57.8% VCT-v2), and Gray Swan (8.5% indirect prompt injection success rate).
- **Non-Examinable Procedures:** Training set composition (contamination audits), Provider Adapter internal source code, and full trajectory logs for mathematical proof discoveries remain closed.

4. Evaluation Against OpenAI's Charter Standard

OpenAI's Charter defines AGI as *"highly autonomous systems that outperform humans at most economically valuable work"*. To operationalize this broad definition into a testable standard, four criteria must be specified:

1. **Economic Scope Threshold ("Most"):** A quantitative threshold defining what percentage of labor hours, wage output, or occupational categories constitutes "most."
2. **Representative Sampling Protocols:** Pre-registered sampling across physical and non-text/non-code enterprise workflows to prevent domain cherry-picking.

3. **Epistemically Fair Human Baselines ("Outperform"):** Standardized comparisons against human experts under domain-calibrated conditions without requiring identical computing bounds.
4. **Autonomy Boundaries ("Highly Autonomous"):** Explicit limits defining permissible human prompt steering, context management, and external tool scaffolding.

The central weakness in treating Astra's release as an established AGI finding is the **missing connection** between disclosed evaluations and the Charter standard: disclosures demonstrate specialized technical strength but contain insufficient evidence for broad, highly autonomous economic coverage.

5. Formal Falsifiability & Verification Matrix

To ensure scientific rigor, we formulate a formal verification matrix detailing operational falsifiability tests for AGI capability assertions:

AGI Claim Dimension	Operational Falsifiability Test	Empirical Failure Threshold	Observed Status (Astra)
Harness Invariance	Evaluate model weights across neutral vs. proprietary harnesses at matched compute.	Performance drop $> 15\%$ on standardized interfaces.	**Failed:** 35.9-point drop on ARC-AGI-3 (62.7% vs 98.6%).
Domain Transfer	Evaluate zero-shot performance on newly released, post-cutoff benchmarks.	$> 30\%$ performance degradation relative to pre-cutoff benchmarks.	**Unverified:** ExploitBench dropped from 100% to 31.7%–39.0%.
Economic Coverage	Representative sampling across SOC occupational categories.	Failure to cover $>$ 50% of non-code/non-text labor tasks.	**Insufficient Evidence:** Disclosures focus on code/text tasks.

AGI Claim Dimension	Operational Falsifiability Test	Empirical Failure Threshold	Observed Status (Astra)
Autonomous Execution	Measure human intervention & prompt correction rates on complex workflows.	Human steering required in $> 20\%$ of multi-turn steps.	**Unverified:** Multi-turn human steering logs remain undisclosed.

6. Master Epistemological Syllogism & Conclusion

Definitive Epistemological Syllogism

Premise 1 (Standard): Presenting an AI system's performance as an established scientific AGI finding requires a clearly specified operational standard—defining economic coverage, fair human baselines, and autonomy conditions—supported by disclosures permitting independent audit.

Premise 2 (Mathematical Bounds): Fixed-parameter autoregressive models M_θ are structurally bounded by $\text{SPACE}\big(O(N)\big) \subsetneq \mathcal{R}$ and the No Free Lunch theorem, restricting their execution to statistical density estimation over $\mathcal{D}_{\text{train}}$ rather than universal computation.

Premise 3 (Assessment): The disclosures examined for GPT-6 Astra provide evidence of high capability in specialized domains, but contain insufficient evidence for the claimed scope—failing to connect the system to highly autonomous human-outperformance across most economically valuable work.

Conclusion: The disclosures examined do not establish that Astra meets OpenAI's Charter standard. Its AGI designation remains **unverified on that record**, with the burden of substantiation remaining strictly on the claimant.

6.1 Summary of Analytical Agreement

Agreement on this audit represents **analytical agreement** regarding logical consistency, burden-of-proof allocation, mathematical bounds, and evidentiary limits. It does not constitute independent empirical verification of undisclosed internal weights, nor does it represent a physical disproof of underlying capabilities. OpenAI’s AGI designation requires evidence connecting the evaluated system to highly autonomous human-outperformance across most economically valuable work. The disclosures examined do not establish that connection; therefore, the designation remains **unverified on that record**.

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